Handling of Confounding and Selection Bias: Propensity Scores

Pràctica 4

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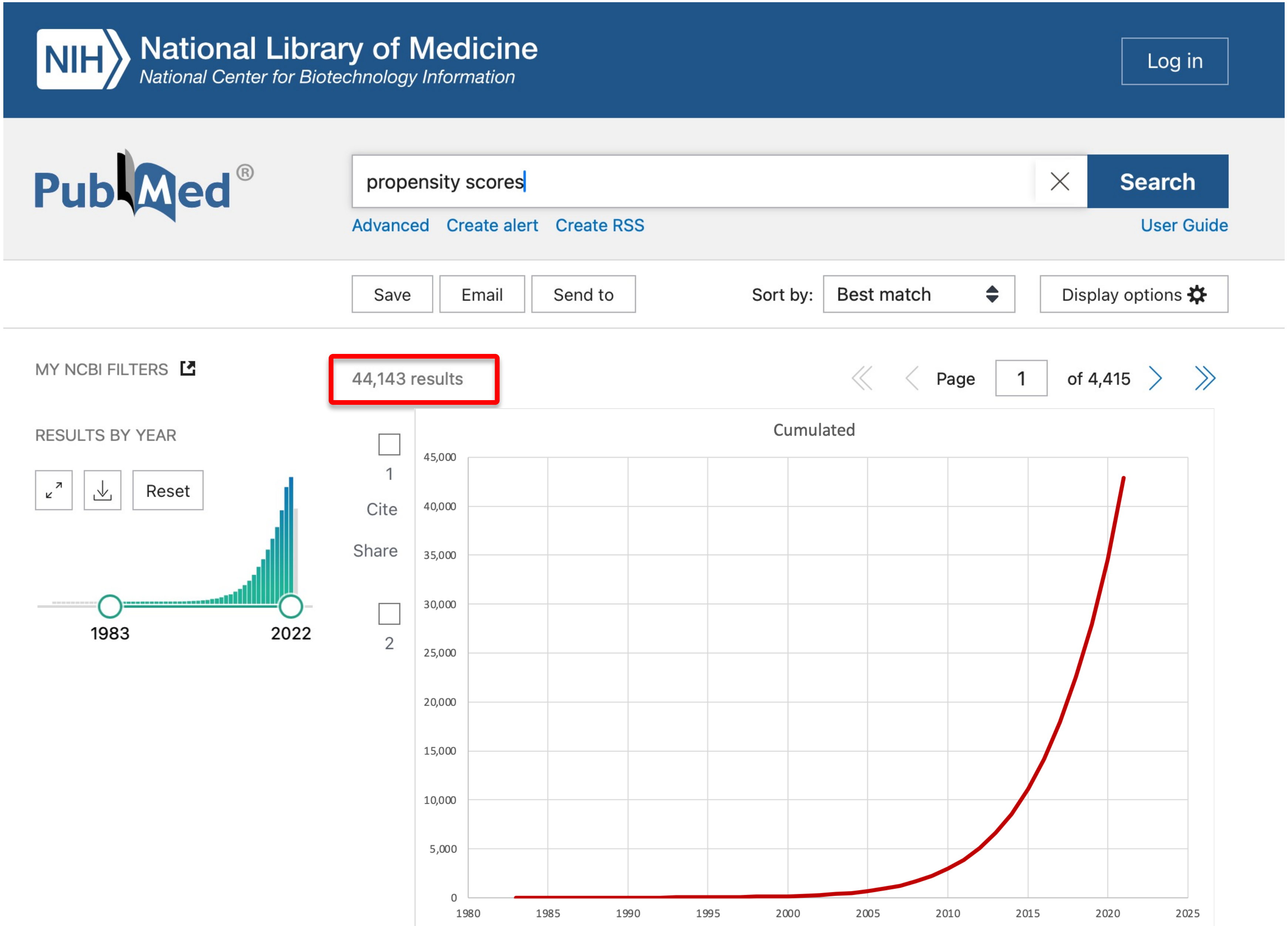
PROPENSITY SCORES



Rosenbaum PR and Rubin DB. Biometrika 1983;70:41–55

Confounding

MI	Coffee No	Coffee Yes
Yes	10	40
No	490	460
	2.0%	**8.0%**

d=6%

Non-Smokers

MI	Coffee No	Coffee Yes
Yes	8	7
No	472	413
	1.7%	**1.7%**

d=0%

Smokers

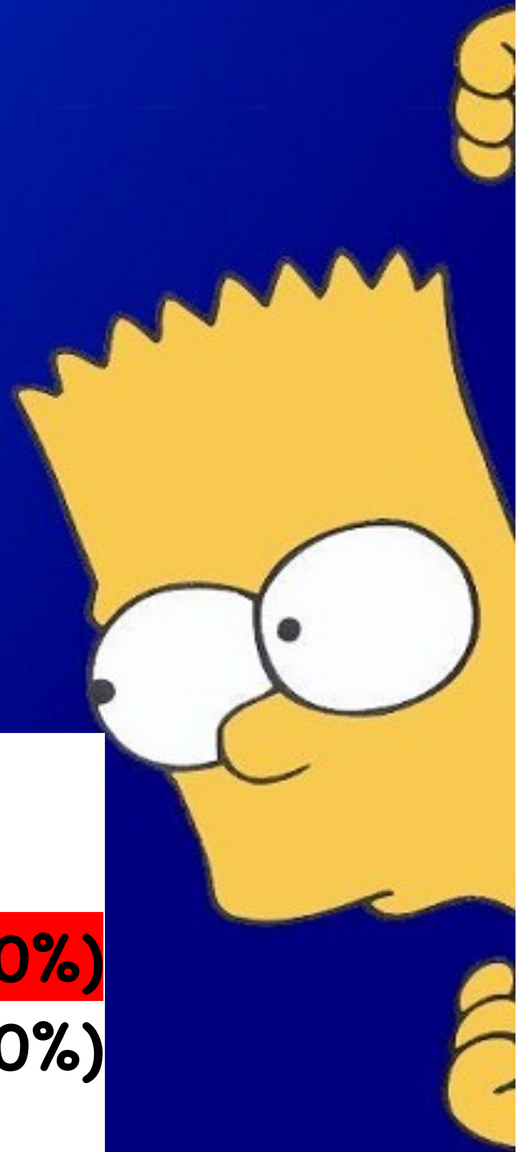
MI	Coffee No	Coffee Yes
Yes	2	7
No	18	73
	10.0%	**8.8%**

d ≈ 0%

Simpson's Paradox

Simpson EH. The interpretation of interaction in contingency tables. J R Stat Soc Series B Stat Methodol 1951; 13: 238–241.

		Experimental n (%)		Control n (%)	
ALL	Succes	70	(70%)	60	(60%)
	Failure	30	(30%)	40	(40%)
		100		*100*	



Simpson's Paradox cont.

		Experimental n (%)	Control n (%)
MALE	Succes	10 (33%)	24 (40%)
	Failure	20 (67%)	36 (60%)
		30	*60*
FEMALE	Succes	60 (86%)	36 (90%)
	Failure	10 (14%)	4 (10%)
		70	*40*

		Experimental n (%)	Control n (%)
ALL	Succes	70 (70%)	60 (60%)
	Failure	30 (30%)	40 (40%)
		100	*100*

Why Randomise

"To eliminate biases that may lead to systematic differences between treatment groups"

Altman DG, Bland JM. BMJ. 1999;318(7192):1209

Randomisation

- Randomization tends to produce study groups that are:
  - Comparable with respect to known **AND unknown** (or unmeasured) risk factors
  - Removes investigator bias in the allocation and treatment of patients
  - Guarantees statistical tests will have valid significance levels.
- Random allocation facilitates blinding

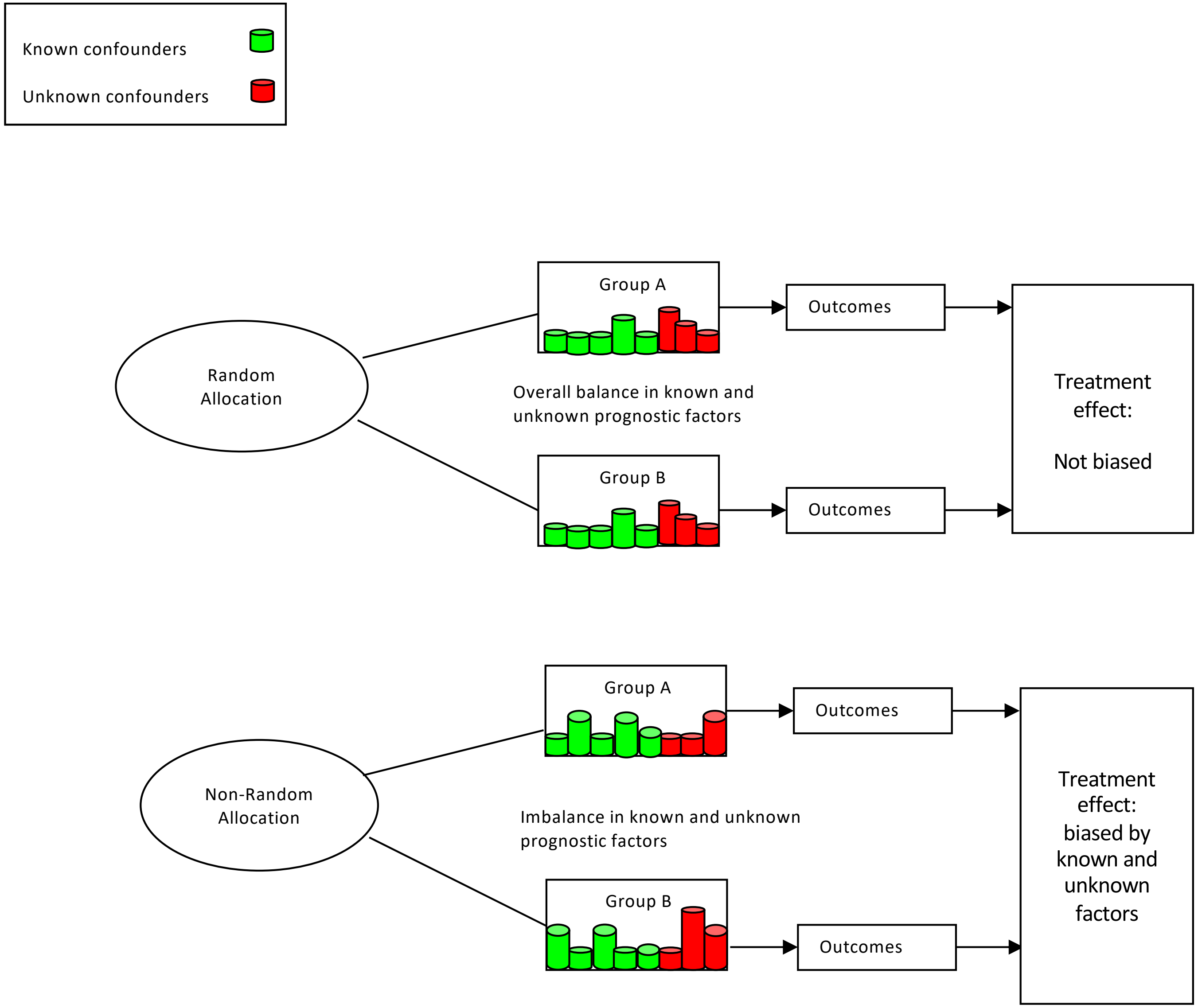


Fig. 1 Differential role of known and unknown confounders in observational studies and in randomized clinical trials. The distribution of unknown confounders is undetermined, but while they are on average balanced in randomized clinical trials, they are not in observational studies.

Torres F et al. Semin Liver Dis. 2017 Aug;37(3):275-286

Assessment of balance

- Statistical significance
  - common, but not recommended

- Standardized difference
  - Strongly preferable

Significance depends on sample size

Characteristic	Treatment A (n = 300)	Treatment B (n = 300)	
Gender			
- Male	63 (21%)	75 (25%)	
- Female	237 (79%)	225 (75%)	
Age (yrs)	49.6 (13.10)	49.9 (13.22)	
Weight (kg)	73.9 (18.95)	75.0 (17.53)	
Height (cm)	164.0 (9.60)	163.3 (9.93)	
BMI	27.5 (6.56)	27.8 (5.94)	
Duration of RA	0.4 (0.48)	0.5 (0.48)	
# of previous DMARDS	0.2 (0.41)	0.2 (0.49)	
Oral corticosteroid use			**0.041**
- No	205 (68%)	180 (60%)	
- yes	95 (32%)	120 (40%)	
Baseline rheumatoid factor			
- positive	273 (91%)	270 (90%)	
- negative	27 (9%)	30 (10%)	
Baseline anti-CCP antibody			
- positive	261 (87%)	358 (86%)	
- negative	39 (13%)	42 (14%)	
Baseline DAS28 Score	6.6 (0.99)	6.7 (0.99)	

Characteristic	Treatment A (n = 250)	Treatment B (n = 250)	
Gender			
- Male	53 (21%)	63 (25%)	
- Female	197 (79%)	187 (75%)	
Age (yrs)	49.6 (13.10)	49.9 (13.22)	
Weight (kg)	73.9 (18.95)	75.0 (17.53)	
Height (cm)	164.0 (9.60)	163.3 (9.93)	
BMI	27.5 (6.56)	27.8 (5.94)	
Duration of RA	0.4 (0.48)	0.5 (0.48)	
# of previous DMARDS	0.2 (0.41)	0.2 (0.49)	
Oral corticosteroid use			
- No	170 (68%)	150 (60%)	**0.077**
- yes	80(32%)	100 (40%)	
Baseline rheumatoid factor			
- positive	228 (91%)	225 (90%)	
- negative	22 (9%)	25 (10%)	
Baseline anti-CCP antibody			
- positive	218 (87%)	215 (86%)	
- negative	32 (13%)	35 (14%)	
Baseline DAS28 Score	6.6 (0.99)	6.7 (0.99)	

Standardized differences for comparing means and prevalences between groups

$$d = \frac{(\bar{x}_{\text{treatment}} - \bar{x}_{\text{control}})}{\sqrt{\dfrac{s^2_{\text{treatment}} + s^2_{\text{control}}}{2}}}$$

d = Difference / SD

$$d = \frac{(\hat{p}_{\text{treatment}} - \hat{p}_{\text{control}})}{\sqrt{\dfrac{\hat{p}_{\text{treatment}}(1 - \hat{p}_{\text{treatment}}) + \hat{p}_{\text{control}}(1 - \hat{p}_{\text{control}})}{2}}}$$

Assessment of balance

Austin PC.
Multivariate Behavioral Research, 46:119–151, 2011

TABLE 1
Baseline Characteristics of the Study Sample

Variable	No Smoking Cessation Counseling (N = 754)	Smoking Cessation Counseling (N = 1,588)	Overall Sample (N = 2,342)	Standardized Difference of the Mean	p Value
		Demographic Characteristics			
Age	60.48 ± 13.26	56.24 ± 11.26	57.61 ± 12.10	0.35	< .001
Female	220 (29.2%)	397 (25.0%)	617 (26.3%)	0.09	.032
		Presenting Signs and Symptoms			
Acute pulmonary edema	34 (4.5%)	48 (3.0%)	82 (3.5%)	0.08	.067
		Vital Signs on Admission			
Systolic blood pressure	146.99 ± 31.82	146.93 ± 29.92	146.95 ± 30.53	0.00	.966
Diastolic blood pressure	84.81 ± 18.99	85.84 ± 18.51	85.50 ± 18.67	0.06	.213
Heart rate	83.28 ± 22.75	81.10 ± 22.54	81.80 ± 22.63	0.10	.029
Respiratory rate	21.18 ± 5.75	20.18 ± 4.64	20.50 ± 5.05	0.20	< .001
		Classic Cardiac Risk Factors			
Diabetes	179 (23.7%)	260 (16.4%)	439 (18.7%)	0.19	< .001
Hyperlipidemia	238 (31.6%)	539 (33.9%)	777 (33.2%)	0.05	.254
Hypertension	295 (39.1%)	541 (34.1%)	836 (35.7%)	0.11	.017
Family history of coronary artery disease	253 (33.6%)	754 (47.5%)	1,007 (43.0%)	0.28	< .001

Assessment of balance

Austin PC.
Multivariate Behavioral Research, 46:119–151, 2011

Variable	*No Smoking Cessation Counseling (N = 682)*	*Smoking Cessation Counseling (N = 682)*	*Standardized Difference of the Mean*	*Standardized Difference of the Mean*	*p Value*
Demographic Characteristics					
Age	59.3 ± 11.8	59.5 ± 13.1	0.012	0.35	< .001
Female	176 (25.8%)	188 (27.6%)	0.040	0.09	.032
Presenting Signs and Symptoms					
Acute pulmonary edema	28 (4.1%)	29 (4.3%)	0.007	0.08	.067
Vital Signs on Admission					
Systolic blood pressure	148.8 ± 31.2	147.7 ± 31.2	0.036	0.00	.966
Diastolic blood pressure	86.4 ± 18.9	85.7 ± 18.5	0.036	0.06	.213
Heart rate	83.2 ± 24.4	82.5 ± 21.9	0.029	0.10	.029
Respiratory rate	20.9 ± 5.3	21.0 ± 5.7	0.027	0.20	< .001
Classic Cardiac Risk Factors					
Diabetes	149 (21.8%)	148 (21.7%)	0.004	0.19	< .001
Hyperlipidemia	217 (31.8%)	218 (32.0%)	0.003	0.05	.254
Hypertension	276 (40.5%)	260 (38.1%)	0.048	0.11	.017
Family history of coronary artery disease	229 (33.6%)	244 (35.8%)	0.046	0.28	< .001

Assessment of Variability

TABLE 3
Baseline Characteristics of Treated and Untreated Participants in the Final Propensity Score Matched Sample

Variable	*No Smoking Cessation Counseling (N = 646)*	*Smoking Cessation Counseling (N = 646)*	*Standardized Difference of the Mean*	*Variance Ratio*
Demographic Characteristics				
Age	59.1 ± 12.4	58.7 ± 12.4	0.026	0.998
Female	178 (27.6%)	175 (27.1%)	0.010	0.989
Presenting Signs and Symptoms				
Acute pulmonary edema	31 (4.8%)	25 (3.9%)	0.046	0.814
Vital Signs on Admission				
Systolic blood pressure	147.7 ± 30.5	147.8 ± 31.3	0.002	1.055
Diastolic blood pressure	85.5 ± 17.7	85.7 ± 18.5	0.015	1.092
Heart rate	82.1 ± 22.4	82.4 ± 22.3	0.013	0.993
Respiratory rate	20.8 ± 5.5	20.9 ± 5.7	0.012	1.046

Austin PC. Multivariate Behav Res. 2011;46(1):119-151

Assessment of Variability

Table IV. Variances of continuous covariates in treated and untreated subjects in unmatched and matched samples.

Variable	Unmatched sample			Matched sample		
	Variance (untreated subjects)	Variance (treated subjects)	Ratio: treated to untreated variances	Variance (untreated subjects)	Variance (treated subjects)	Ratio: treated to untreated variances
Age	191.71	153.54	0.80	174.84	158.08	0.90
Heart rate	590.84	527.08	0.89	508.03	521.28	1.03
Systolic BP	996.68	904.15	0.91	914.95	909.29	0.99
Diastolic BP	346.7	323.95	0.93	327.17	337.04	1.03
Respiratory rate	32.9	22.81	0.69	23.32	22.09	0.95
WBC	23.75	19.51	0.82	17.2	21.64	1.26
Hemoglobin	374.32	286.34	0.76	331.62	288.36	0.87
Sodium	15.37	10.82	0.70	13.96	10.75	0.77
Glucose	26.1	28.05	1.08	26.04	30.77	1.18
Potassium	0.32	0.26	0.82	0.28	0.26	0.94
Creatinine	4283.81	2503.02	0.58	3477.95	2745.08	0.79

Austin PC
Stat Med. 2009;28:3083-107

Criteria for balance

- Differences (SMD)
  - No universal rule
  - **<10%**

Imai K, Van Dyk DA. J Am Stat Assoc 2004;99:854-866
Ali MS et al. J Clin Epidemiol 2015;68:122-131
Ali MS et al. Am J Clin Nutr 2016;104:247-258
Austin P C. *Stat Med 2009;*28:3083-3107

- Ratio of variances
  - Ratio of 1.0 ideal, and <2 acceptable balance

Rosenbaum P R, Rubin DB. *Am Stat 1985;*39:33-38
Rubin DB. *Health Serv. Outcomes Res Methodol 2001;*2:169-188
Ho D et al. *Polit. Anal* 2007;15:199-236
Austin P C. *Stat Med 2009;*28:3083-3107
Linden A, Samuels SJ. *J Eval Clin Pract 2013;*19:968-975

Assessment of balance

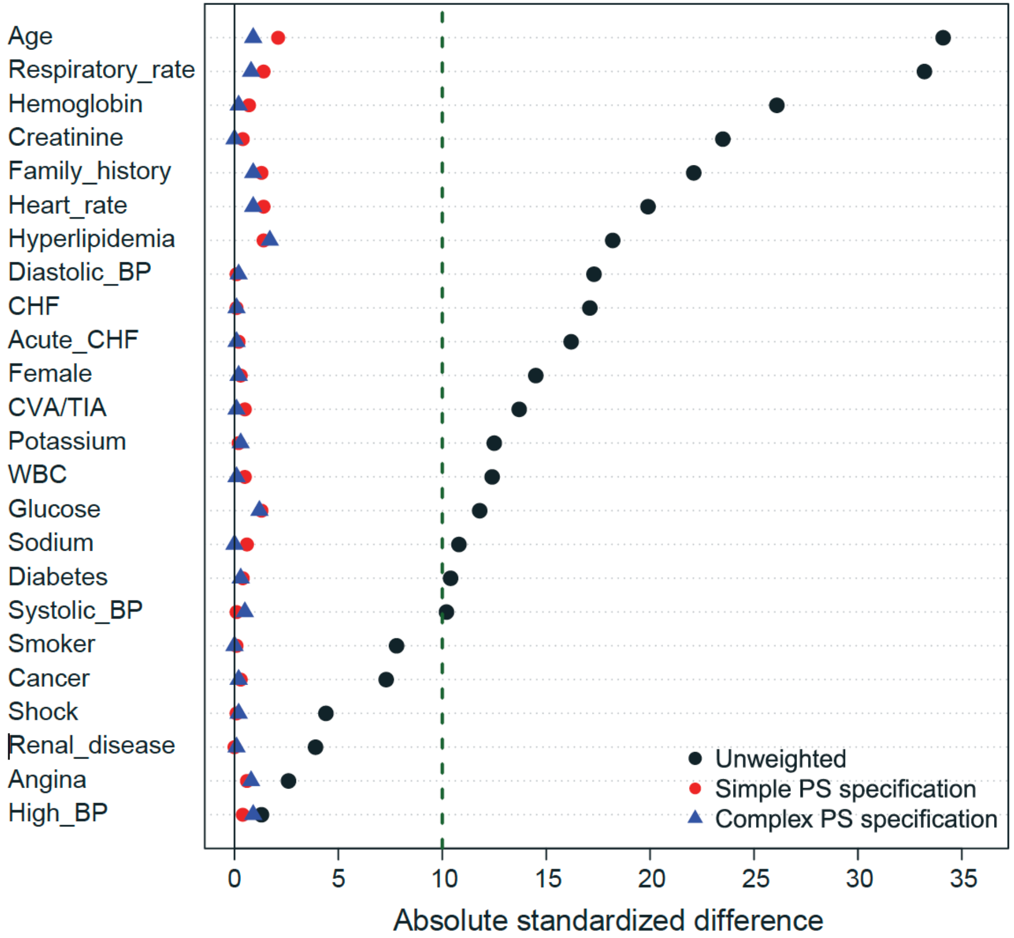


Figure 2. Absolute standardized differences in unweighted and weighted samples.

Austin & Stuart. Stat Med. 2015;34:3661-79

“Grouping variable”

- Case-Control (CC) study:
  - Designed group is **CC** (the **outcome** result)

  CC = **Exposure** + $Covar_1$… $Covar_X$

- Cohort Study:
  - Designed group is the **exposure** **(Cohort)**

  Outcome = **Cohort** + $Covar_1$… $Covar_X$

- Both:
  - Adjust by confounders $Covar_1$… $Covar_X$

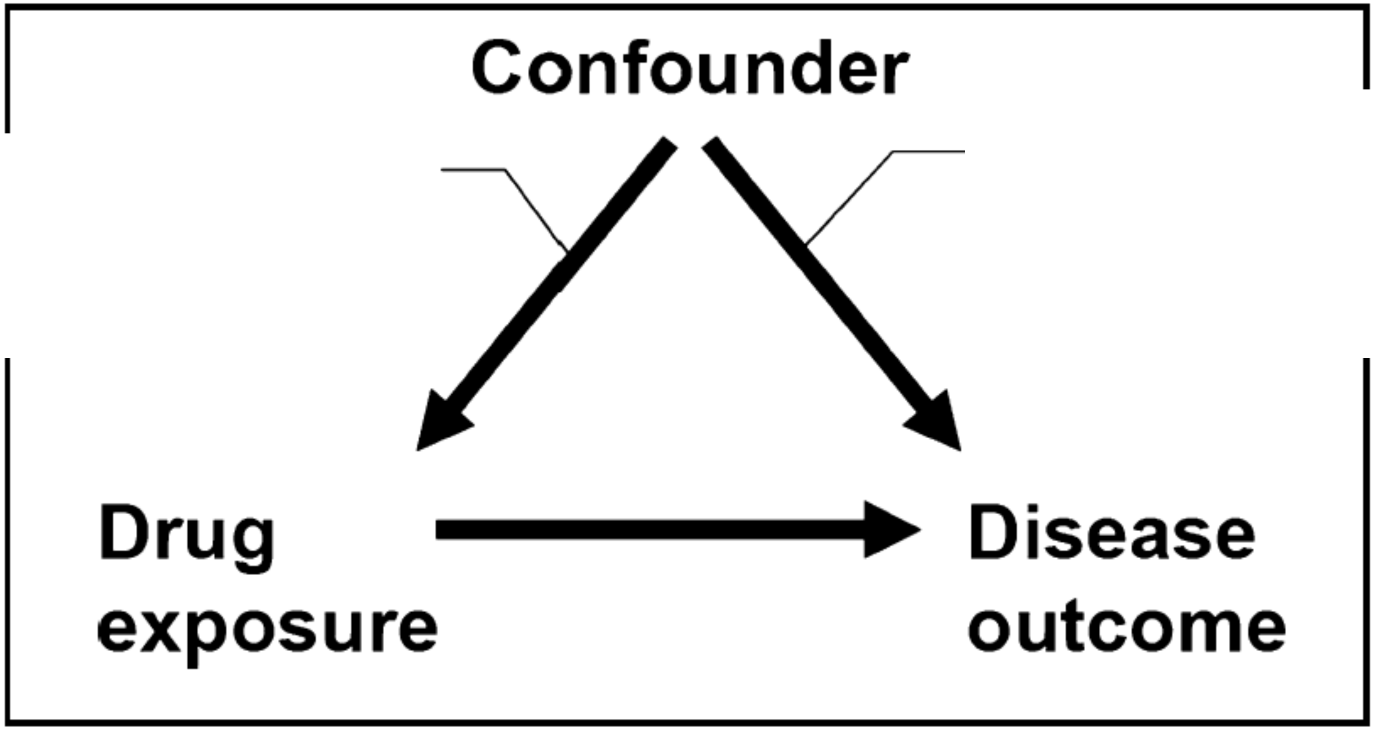
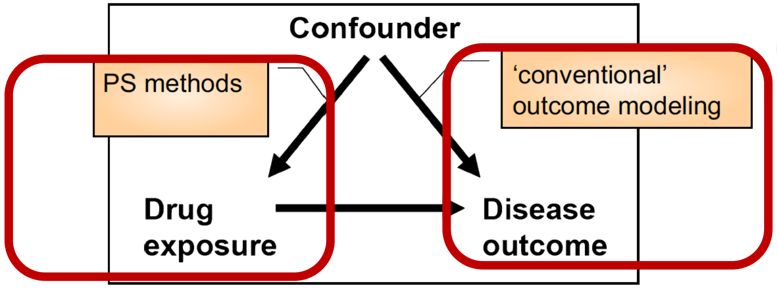


Fig. 1 Causal diagram for confounding. Both arrows need to be present to cause confounding. Removing one of the arrows removes confounding. This leads to essentially two general ways to control for confounding: we can remove the effect of the confounder on treatment (propensity score methods), or we can condition on the effect of the confounder on the outcome (outcome modelling). Both are dependent on having a good measure of the confounder available to the researcher.

Stürmer et al. J Intern Med. 2014;275:570-80



Cohort Study:

– Designed group is the **exposure** **(Cohort)**

Outcome = **Cohort** + $\text{Covar}_1 \ldots \text{Covar}_X$

- PS
  - Outcome result initially <u>ignored!</u>
  - Group (<u>**cohort**</u>) predicted from covariates:

    <u>**Cohort**</u> = $\text{Covar}_1 \ldots \text{Covar}_X$
  - Individual probabilities taken from that model
  - $\text{Covar}_1 \ldots \text{Covar}_X$ not used in the final analysis of the outcome (just the PS)

What is a "propensity score"?

- the **probability** that a patient will receive the treatment of interest, based on a series of **covariates**
  - the patient characteristics, the treating physician, and the clinical environment

- frequently estimated by logistic regression

- Replace the collection of confounding covariates with one function of these covariates: the propensity score (**probability**)

Practical application of PS

- Once the value of the PS has been calculated, there are several ways to apply it

1) Stratification

2) Adjustment (regression)

3) IPTW (inverse probability of treatment weighting)

4) Matching

IPTW

- IPTW (*Inverse Probability of Treatment Weighting*).

- Instead of the PS, we will use the calculated <u>weights</u> as a <u>weight</u> the model:

$$W_{exp} = \frac{1}{PS_i} \qquad W_{control} = \frac{1}{1 - PS_i}$$

Un-Stabilised weights

A very large/small weight

- Propensity score **trimming**
  - Exclude extremes 1% or 2.5% or 5%

Stürmer et al. J Intern Med. 2014;275:570-80

- **Stabilized weights:**
  - The numerator of the unstabilized weights, which is 1, is replaced by the proportion of participants who selected the treatment

Thoemmes & Ong. Emerging Adulthood 2016;4:40-59

Average Treatment (AT) estimand

ATE

- **A**verage **T**reatment **E**ffect
- By default in IPTW
- Average effect of moving an entire population from untreated to treated
- When every treatment potentially might be offered to every subject

ATT

- **A**verage **T**reatment on the **T**reated
- By default in Matching
- Average effect of treatment on those subjects who ultimately received the treatment
- When patient's characteristics are more likely to determine the treatment received.

Average Treatment (AT) estimand

ATE

- *IPTW*
  - $T/\mathrm{PS} + (1 - T)/(1 - \mathrm{PS})$
- IPTW-ATT
  - $T + \mathrm{PS}(1-T)/(1-\mathrm{PS})$

ATT

- Matching

Other estimands

Table 2: Target populations, tilting functions, estimands and the corresponding balancing weights for binary treatments in PSweight.

Target population	Tilting function $h(\boldsymbol{x})$	Estimand	Balancing weights (w_1, w_0)
Combined	1	ATE	$\left(\frac{1}{e(\boldsymbol{x})}, \frac{1}{1-e(\boldsymbol{x})}\right)$
Treated	$e(\boldsymbol{x})$	ATT	$\left(1, \frac{e(\boldsymbol{x})}{1-e(\boldsymbol{x})}\right)$
Overlap	$e(\boldsymbol{x})(1-e(\boldsymbol{x}))$	ATO	$(1-e(\boldsymbol{x}), e(\boldsymbol{x}))$
Matching	$\xi_1(\boldsymbol{x})$	ATM	$\left(\frac{\xi_1(\boldsymbol{x})}{e(\boldsymbol{x})}, \frac{\xi_1(\boldsymbol{x})}{1-e(\boldsymbol{x})}\right)$
Entropy	$\xi_2(\boldsymbol{x})$	ATEN	$\left(\frac{\xi_2(\boldsymbol{x})}{e(\boldsymbol{x})}, \frac{\xi_2(\boldsymbol{x})}{1-e(\boldsymbol{x})}\right)$

Notes: $\xi_1(\boldsymbol{x}) = \min\{e(\boldsymbol{x}), 1-e(\boldsymbol{x})\}$ and $\xi_2(\boldsymbol{x}) = -\{e(\boldsymbol{x})\log(e(\boldsymbol{x})) + (1-e(\boldsymbol{x}))\log(1-e(\boldsymbol{x}))\}$.

Zhou 2021.
PSweight: An R Package for Propensity Score Weighting Analysis
https://arxiv.org/pdf/2010.08893.pdf

Criteria for Covariates selection

- …may influence the treatment selection process
- …may induce balance, regardless of
  - their statistical significance
  - or collinearity with other variables in the model
- … include all risk factors and important interaction terms among confounders
- not all covariates are of equal importance:
  - those to likely affect the outcome
- not using the outcome data in the PS process

Austin & Stuart. Stat Med. 2015;34:3661-79; Elze et al. J Am Coll Cardiol. 2017;69:345-357

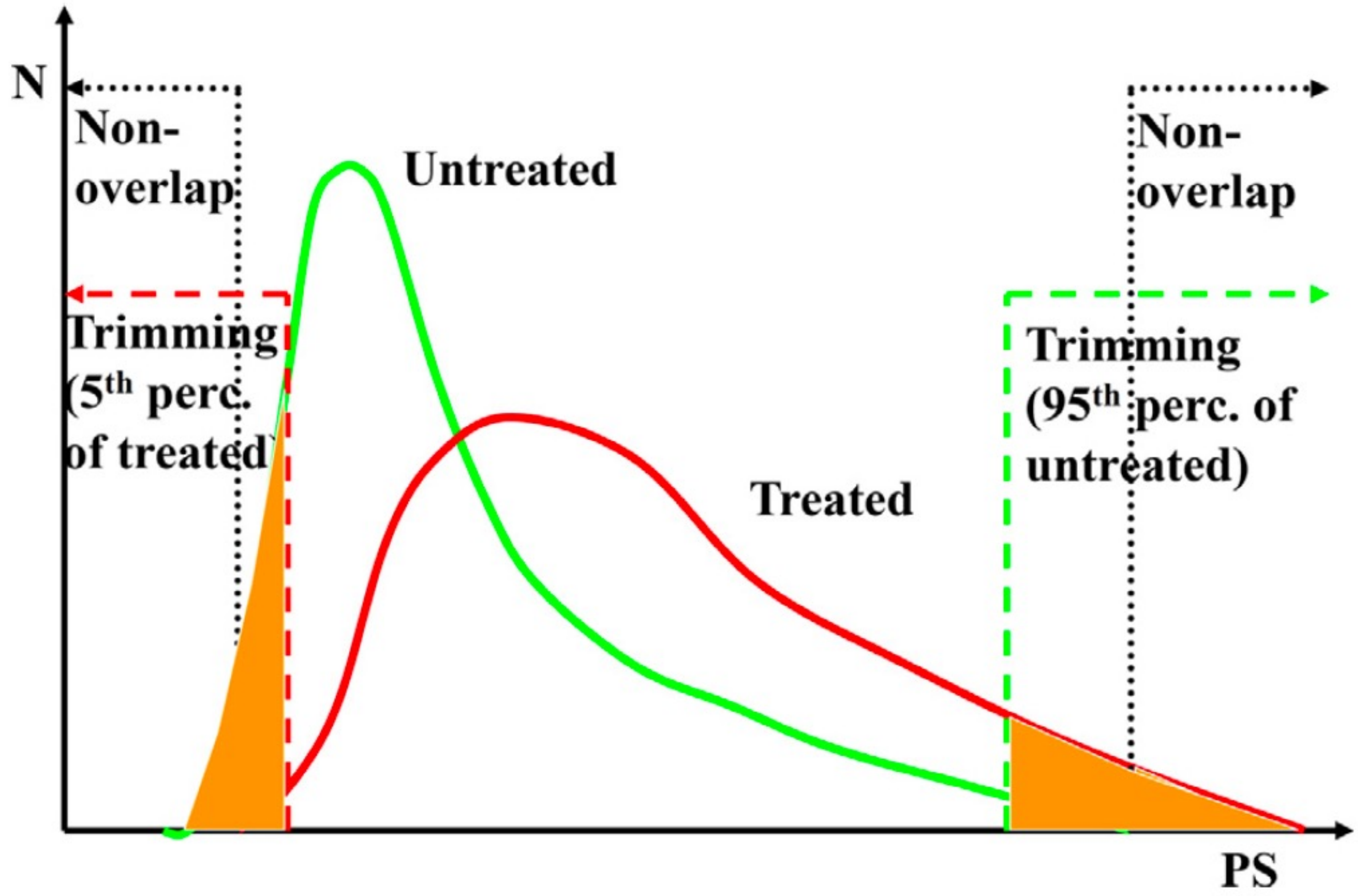


Fig. 8 *Schematic for trimming at the ends of the (overlapping) propensity score distribution.*

Stürmer et al. J Intern Med. 2014;275:570-80

Steps for PS analyses

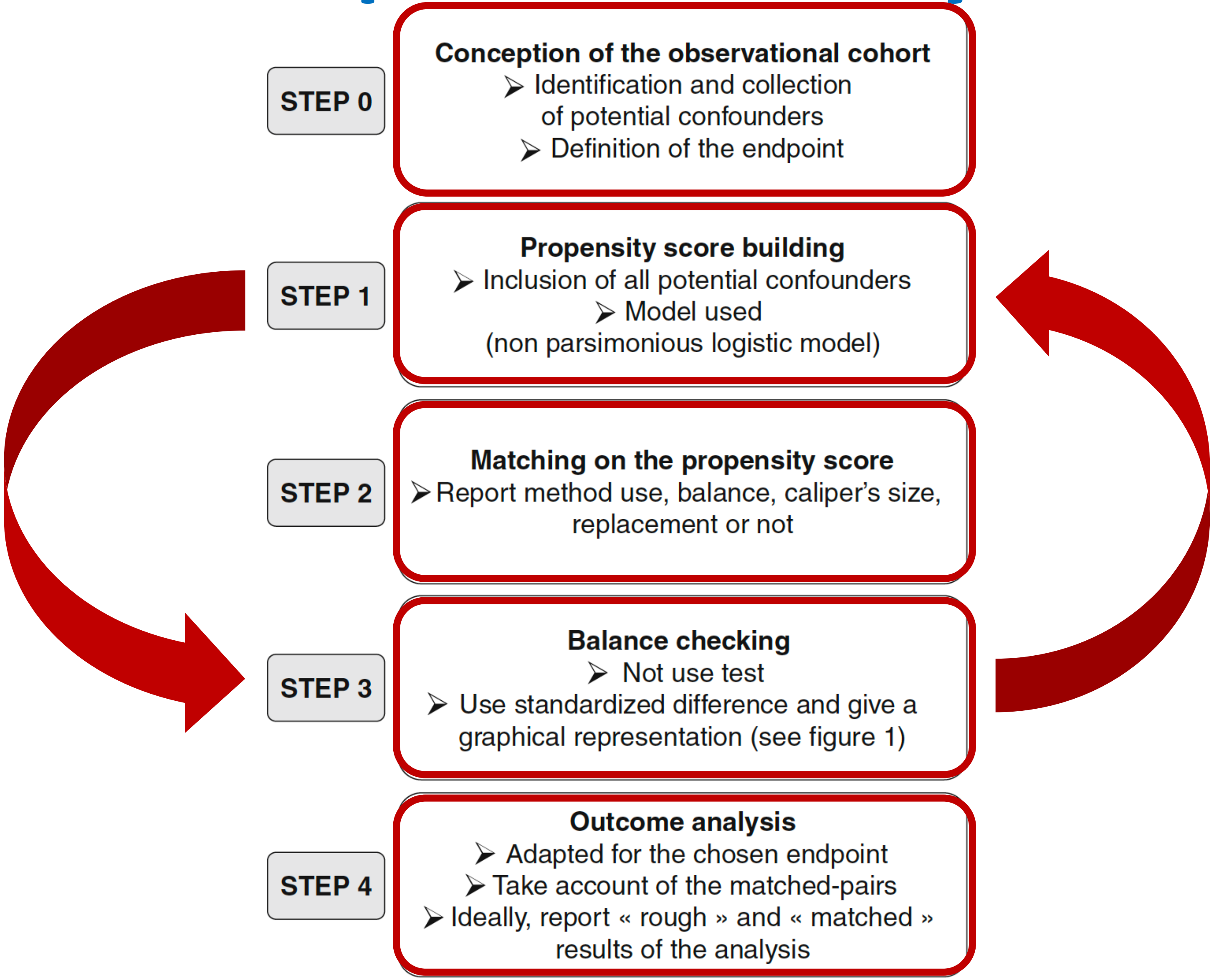
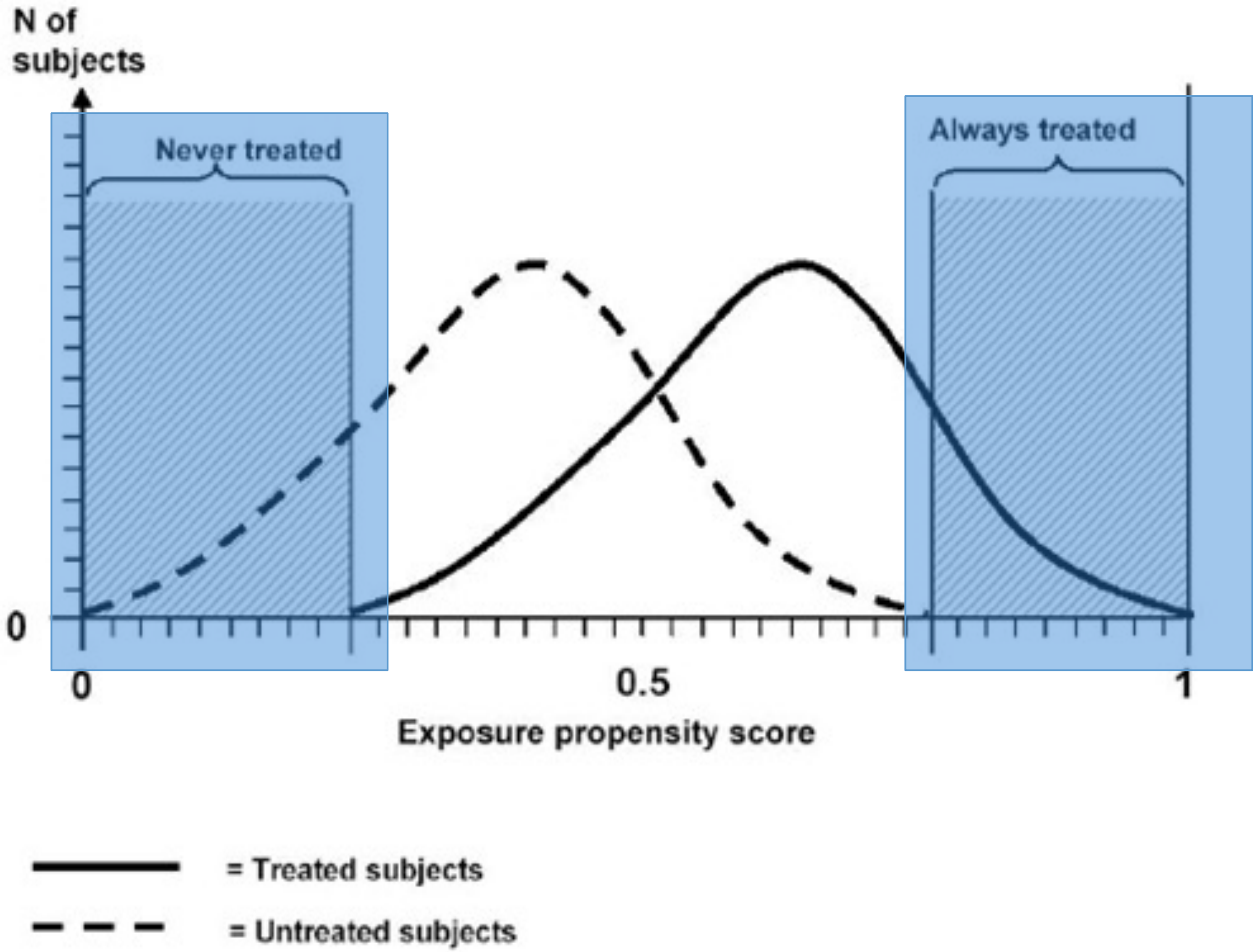
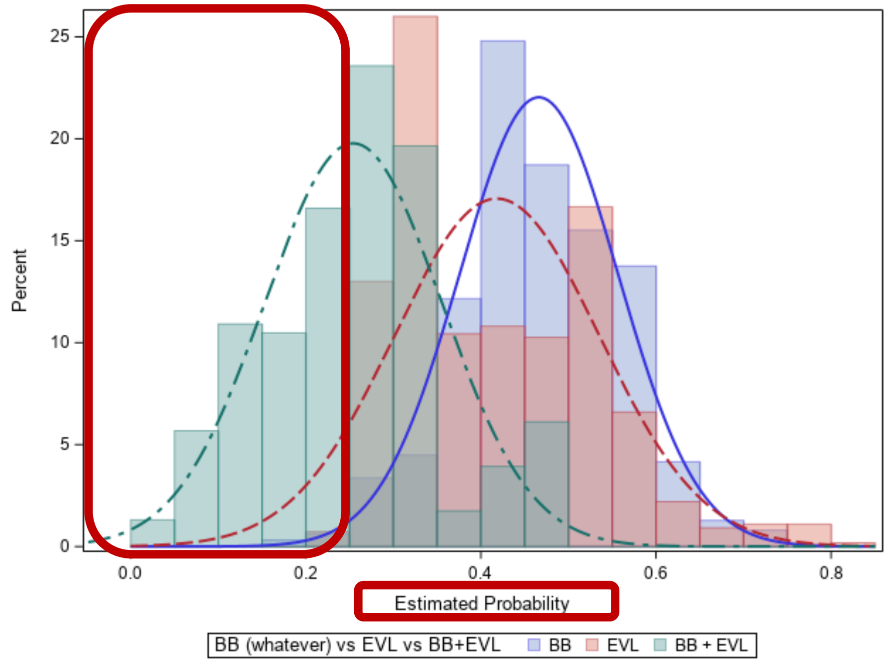


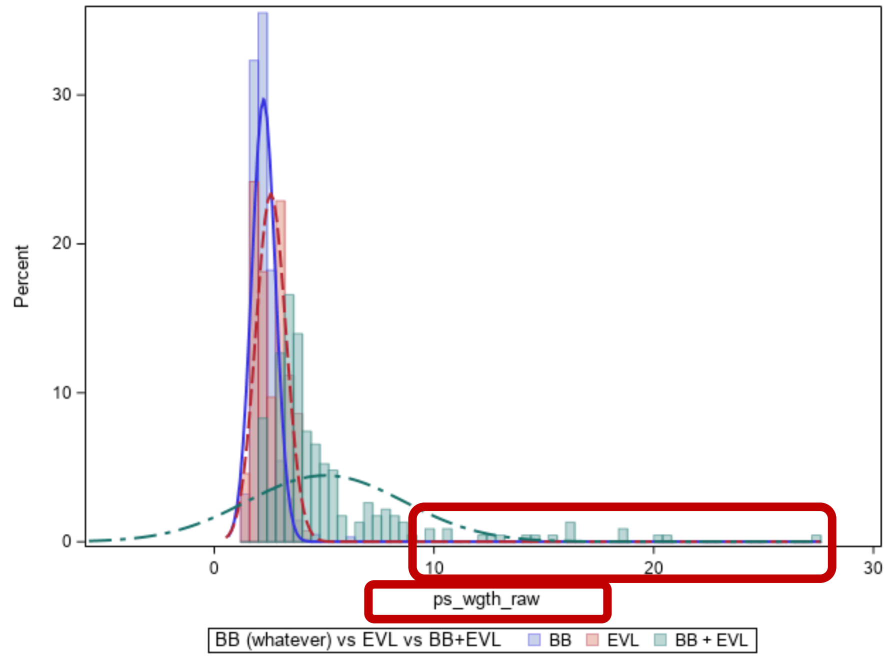
Fig. 3 Steps for planning and analysis of an observational study by means of propensity score methods.

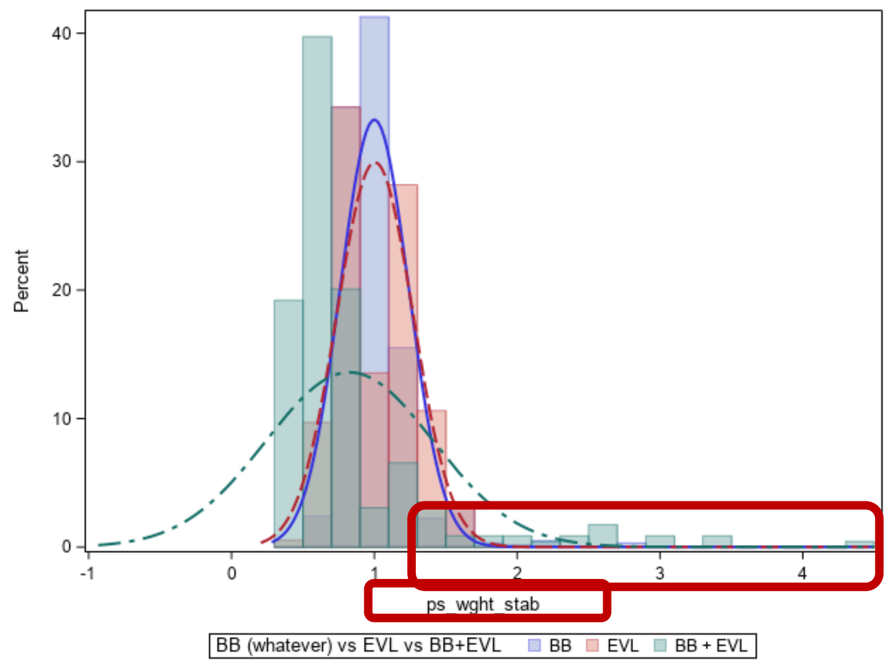
Gayat et al.
Intensive Care Med. 2010;36:1993-2003

Limitaciones del uso del Propensity Score.









Torres F, Ríos J, Saez-Peñataro J, Pontes C.
Is Propensity Score Analysis a Valid Surrogate of Randomization for the Avoidance of Allocation Bias?
Semin Liver Dis. 2017;;37(3):275-286.

Table v**0** to check **baseline** balance T=1 vs T=0
|STD| >10% ?

i=0
to
….

Use (some) **baseline** covariates from Table v**i** with STD>10%
Logistic regression to predict T=1
i=i+1
Model v**i**

Use predicted probabilities from Model v**i**
Calculate stabilised **weights**

Table v**i** with **weights** to check **baseline** balance T=1 vs T=0
|STD| >10% ?

Check weights v**i** (descriptive, figures) reiterate if needed
Use weights v**i** for **final analysis** with **outcome**

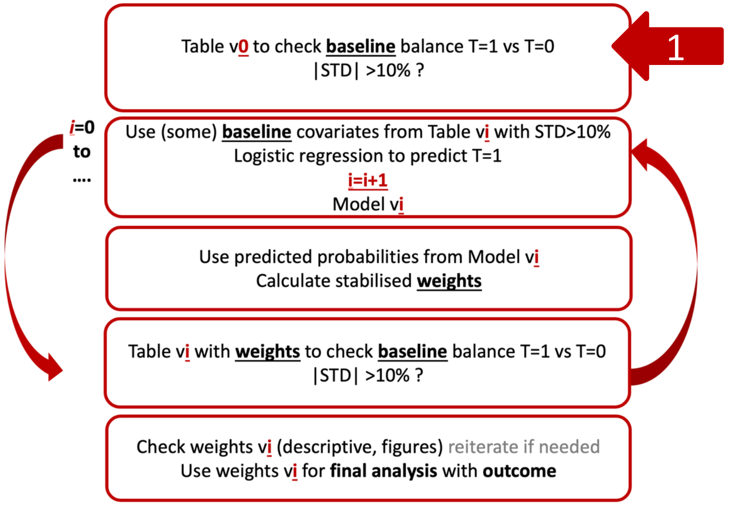


Table 2. Donor- and graft-related characteristics.

Characteristic	Raw analysis			
	NRP (n = 95)	SRR (n = 117)	*p* value	Standardised difference[a]
Age (yr)	53.8 (15.2) 57 (45–65)	54.5 (12.0) 56 (47–64)	0.796	−0.050
Sex male	63 (66.3%)	77 (65.8%)	0.939	0.011
Cause of death				
CVA	42 (44.2%)	49 (41.9%)	0.733	0.047
Anoxic brain injury	38 (40.0%)	47 (40.2%)	0.980	−0.004
Traumatic brain injury	8 (8.4%)	13 (11.1%)	0.514	−0.091
Other	7 (7.4%)	8 (6.8%)	0.881	0.021
ICU stay (days)	9.2 (9.9) 7 (4–12)	9.5 (8.5) 7 (5–11)	0.460	−0.033
Total WIT[b] (min)	19.2 (8.2) 18 (13–23)	23.1 (6.7) 22 (19–26)	<0.001	**−0.515**
Functional WIT[c] (min)	13.3 (5.3) 12 (9–16)	16.1 (5.3) 15 (12–20)	<0.001	**−0.541**
CIT (min)	333.1 (109.8) 315 (265–365)	340.7 (94.2) 340 (285–383)	0.141	−0.075
Preservation solution				
UW or IGL-1	37 (38.9%)	15 (12.8%)	<0.001	**0.625**
HTK	1 (1.1%)	27 (23.1%)	<0.001	**−0.719**
Celsior	57 (60.0%)	75 (64.1%)	0.540	−0.085

Table 3. Recipient- and transplant-related characteristics.

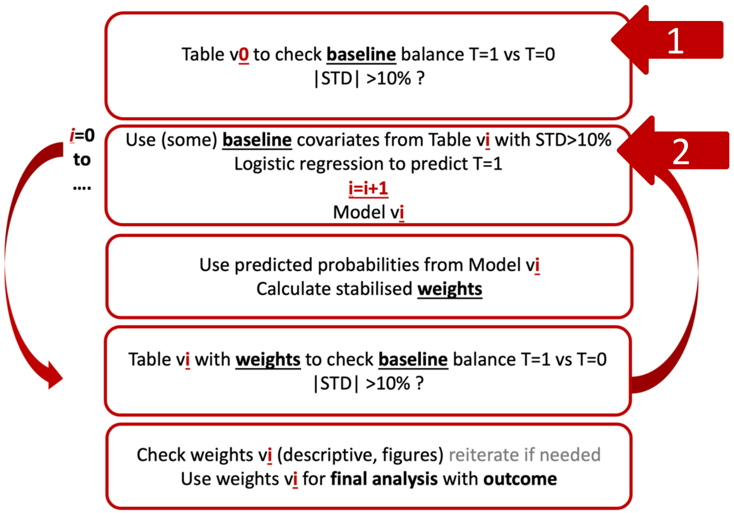
1 =>

Stddiff (no weights!)

library(tableone)

TabNoIPTW <- CreateTableOne(vars =ListOfVars, strata="treat", data=baseline….)

Hepatocellular carcinoma	35 (36.8%)	38 (32.5%)	0.506	0.092
Retransplantation or fulminant liver failure	2 (2.1%)	2 (1.7%)	0.833	0.029
Other	5 (5.3%)	2 (1.7%)	0.150	**0.195**



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ICU stay (days)	9.2 (9.9)	9.5 (8.5)	0.460	−0.033
	7 (4–12)	7 (5–11)		
Total WIT[b] (min)	19.2 (8.2)	23.1 (6.7)	<0.001	**−0.515**
	18 (13–23)	22 (19–26)		
Functional WIT[c] (min)	13.3 (5.3)	16.1 (5.3)	<0.001	**−0.541**
	12 (9–16)	15 (12–20)		

1 =>

Stddiff (no weights!)

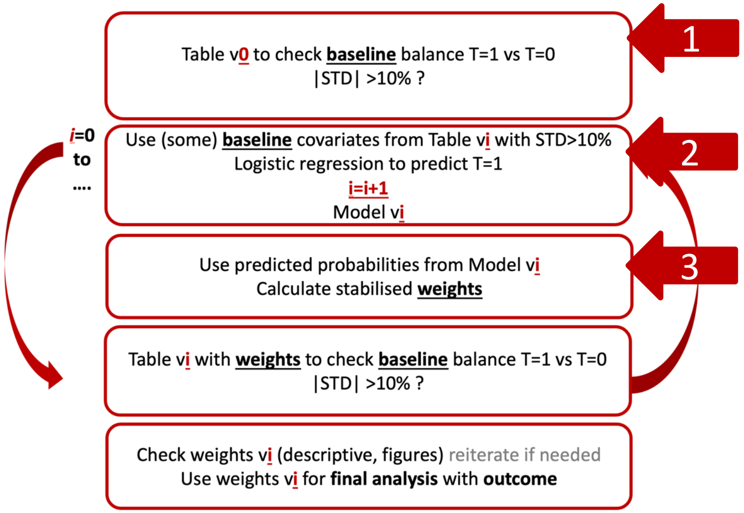
library(tableone)

TabNoIPTW <- CreateTableOne(vars =ListOfVars, strata="treat", data=baseline....)

2 =>

m**i** <- glm(treat ~CoVar1+ … CoVar2, data=baseline, family = binomial(link = 'logit'))

[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
	15 (11–19)	13 (9–18)		
High-volume transplant centre[c]	69 (72.6%)	88 (75.2%)	0.670	−0.059
Transplant indication				
Cirrhosis	53 (55.8%)	75 (64.1%)	0.218	**−0.170**
Hepatocellular carcinoma	35 (36.8%)	38 (32.5%)	0.506	0.092
Retransplantation or fulminant liver failure	2 (2.1%)	2 (1.7%)	0.833	0.029
Other	5 (5.3%)	2 (1.7%)	0.150	**0.195**

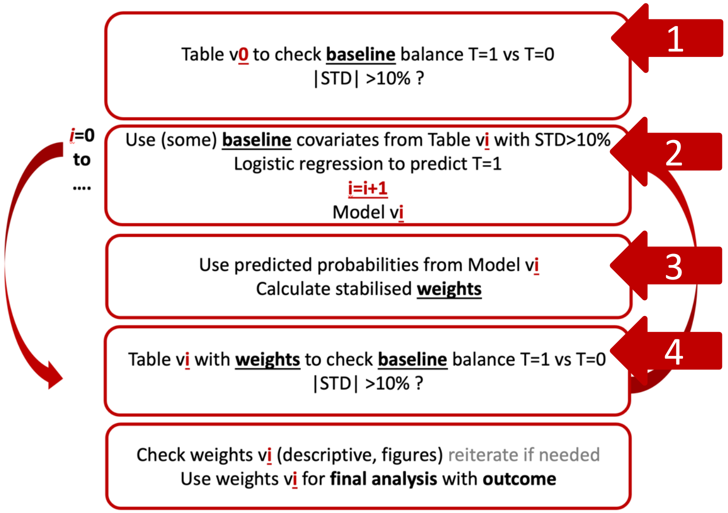


2 =>

```
mi <- glm(treat ~CoVar1+ … CoVar2, data=baseline, family = binomial(link = 'logit'))
```

3 =>

```
baseline <- baseline %>%
  mutate(probi = predict(mi, type = 'response')) %>%
  mutate(IPTWi = 0.3808*treat/probi + 0.6192 *(1-treat)/(1-probi))
```



3 =>

```
baseline <- baseline %>%
  mutate(probi = predict(mi, type = 'response')) %>%
  mutate(IPTWi = 0.3808*treat/probi  +  0.6192*(1-treat)/(1-probi))
```

4 =>

```
## Weighted data
baselineIPTWi <- svydesign(ids=~1, data=baseline, weights=~IPTWi)
## Construct a table (This is a bit slow)
TabIPTWi <-svyCreateTableOne(vars=ListOfVars, strata="treat", data=baselineIPTWi)
```

Recommendations

- Include only the worst unbalanced CoVars in step 1
- Then include a new unbalanced one in each step
- Variables are not independent: a new CoVar entry implies changes in others
- Do not use p-values criteria for variable selection => STD !!!

Recommendations

- Continuous variables:
  - First include single terms, try linear term with no transformation and then add additional terms (X^2, X^3, $1/X$, $\ln(X)$, $\mathrm{sqrt}(X)$ etc…
- Categorical less options:
  - Included/excluded
  - Re-categorise or use dummies
- Try interaction in later stages

Propensity scores approaches with R

wtd.t.test
weighted.mean

…

```
glm(Outcume ~ treat, weights = IPTW, data =data, family = binomial("logit"))
```

```
Coxph(formula, data=, weights=IPTW, subset,
    na.action, init, control,
    ties=c("efron","breslow","exact"),
    singular.ok=TRUE, robust,
    model=FALSE, x=FALSE, y=TRUE, tt, method=ties,
    id, cluster, istate, statedata, nocenter=c(-1, 0, 1), ...)
```